



# ADVANCING RESILIENT CRITICAL MINERAL SUPPLY CHAINS IN INDONESIA

## A TRIUMVIRATE APPROACH

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**E**nsuring reliable and secure access to critical minerals is a top priority for the United States as it seeks to safeguard its position in frontier technologies such as solar panels, electric vehicles, batteries, and semiconductors. Disruptions to mineral supplies could undermine U.S. competitiveness in these strategic industries, and global supply chains for many critical minerals currently run through China.

China's investment in Indonesia followed Beijing's "going out" strategy—a state-backed push for Chinese firms to invest abroad. The strategy's goal was to secure supply chains, expand global influence, and support domestic economic growth by investing in natural resources, infrastructure, and strategic industries abroad.

The national security implications of this became evident in December 2024 when China banned exports of gallium and germanium to the United States.<sup>1</sup> The ban of these critical minerals used in the production of semiconductors and other electronics roiled markets and led to shortages, underscoring U.S. vulnerability to supply chain shocks. Although the ban was lifted at the end of 2025 alongside a number of other subsequent export controls, the consequences highlighted for the United States the urgency of securing access to critical minerals through channels that bypass China.

The scale of the challenge is significant. China has spent years consolidating control over chokepoints in supply chains for critical minerals, securing access to raw ores and investing heavily in refining capacity both domestically and abroad. In 2024, China produced 99% of the world's gallium and 44% of its refined copper.<sup>2</sup> What it does not

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produce domestically, it imports through state-owned or state-directed mining companies, including cobalt and copper from the Democratic Republic of Congo and nickel from Indonesia.

The United States has seemingly been slow to recognize the geopolitical value of critical minerals, allowing procurement, investment, and other supply chain decisions to be driven largely by market incentives and the interests of private firms. Addressing the national security implications requires looking beyond commercial considerations and forming partnerships with like-minded allies based on shared geostrategic priorities rather than exclusively market logic.

Current schemes, such as the Section 45X tax credit for critical minerals, incentivize onshoring and domestic production. But a resilient global supply chain is not necessarily one where the United States sits at the center, absorbing raw minerals from other countries and processing them into high-value finished goods. Resilient global supply chains take the form of distributed cooperative networks, where U.S. firms with know-how and technology invest in overseas mining, smelting, and manufacturing through subsidiaries set up in resource-rich countries. This will allow U.S. firms to gain access to scarce resources, while providing attractive benefits to the countries that own and control the primary inputs.

China may hold a leading position, but the United States can still build alternative, resilient supply chains by partnering with resource-rich states in the Asia-Pacific, including Indonesia and Australia. The recent U.S.-Australia critical minerals framework marks an important step in this direction.<sup>3</sup> Incorporating Indonesia would strengthen it further.

The key question is how such frameworks can be structured to benefit all parties at a time when many countries—including the United States and Indonesia—are considering how to shape global supply chains to advance national interests and expand domestic production. The following sections will examine this question as it relates to prospects

for U.S.-Indonesian cooperation on critical minerals for batteries (nickel and copper) and semiconductors (germanium and gallium).

## Critical Minerals for Batteries and Semiconductors

Indonesia plays a central role in supply chains of critical minerals necessary for the production of batteries and semiconductors, two technologies with widespread applications. Indonesia held 42% of global nickel ore reserves in 2024, and its mines accounted for 59% of global production.<sup>4</sup> Nickel is used to make lithium-ion batteries and is thus central to clean energy supply chains. Historically, large foreign mining firms extracted Indonesian nickel ore and exported it for processing outside Indonesia. Seeking to retain more value domestically amid heightened global competition over control of critical mineral resources, Indonesia in 2009 passed a law requiring that ore be processed in domestic smelters and in 2020 banned nickel ore exports to enforce compliance. This policy succeeded in driving massive investment into Indonesia's downstream nickel industry.<sup>5</sup>

Billions of dollars—primarily from Chinese firms—flowed into smelters and industrial parks in nickel-rich regions of Indonesia. In addition to expanded mineral-processing facilities, factories to produce batteries and electric vehicles are also now in various stages of development, backed mainly by Chinese and South Korean investors. The impact of this downstreaming policy can also be seen in Indonesia's rapidly expanded output of refined products such as ferronickel and nickel matte. In 2017, Indonesia exported just under \$2 billion worth of these products. By 2023, it was exporting nearly \$21 billion, with the majority of exports going to China.<sup>6</sup>

Indonesian nickel production increased so rapidly the market is now oversupplied and prices have fallen. The Gunbuster nickel smelter was forced to ramp down recently after its parent company in China declared bankruptcy, partly due to falling

nickel prices.<sup>7</sup> This could be interpreted as evidence the policy is not working as intended. But that conclusion hinges on the assumption that Indonesia's nickel downstreaming is driven solely by commercial logic, and that success or failure should be judged using metrics like profit and loss, or prices determined by markets. Viewed instead through a geopolitical lens, this trend shows that Indonesia and China are working toward mutually beneficial objectives. Indonesia leveraged its dominant nickel resources to expand its footprint in clean energy supply chains, prioritizing its interest in downstreaming even when conditions were not commercially optimal. China sought long-term access to nickel for next-generation technologies and was willing to invest, even with low or negative margins, because securing supply has value beyond short-term commercial returns.

The extent to which downstreaming is resulting in the creation of long-term value for Indonesia, in the form of technology transfers and upskilling, remains to be seen. Substantial environmental and social costs are also being incurred as part of this policy. Whether the benefits outweigh the costs remains a matter of intense debate. What is indisputable is that downstreaming has helped Indonesia insert itself into the global conversation on critical minerals in a way that was not possible before.

Indonesia now seeks to replicate this experience with copper smelters and downstream industries, having enacted export bans on unprocessed copper and bauxite in 2023.<sup>8</sup> Indonesian mines accounted for 5% of global copper production in 2024, but the country's downstream industries remain underdeveloped. The United States already has an established private-sector presence here through Freeport-McMoRan, which has operated the Grasberg mine in Papua for decades. If securing reliable access to non-Chinese copper is a strategic priority, the United States is well placed to help Indonesia realize its downstream ambitions in exchange for priority access to refined copper products.

Gallium and germanium are also considered critical minerals, though they are not mined like nickel or copper. Gallium is a byproduct of bauxite and zinc processing; germanium can be extracted from coal ash. Their market size is small in dollar terms, but they are important components in semiconductors and other electronics. China currently dominates global production of both minerals—mainly because manufacturing them is uneconomical at scale, and other producers cannot compete on a purely market basis with subsidized Chinese firms. U.S. companies have shown little interest in producing gallium and germanium because they cannot do so profitably. However, considering the role of these minerals in semiconductor supply chains, they have strategic value that goes beyond their market price.

Suitable candidates for producing gallium and germanium outside China would be countries rich in bauxite and coal that place a premium on noncommercial objectives, such as improving their position in global supply chains. Indonesia is such a country. It contains 10% of global bauxite reserves and is one of the largest exporters of coal in the world.<sup>9</sup> Given that China's recent export bans reminded the United States about the vulnerability of its supply chains for critical minerals, expanding partnerships with a friendly country like Indonesia that possesses potentially favorable conditions for producing these minerals with U.S. assistance is worth consideration.

## **Building Critical Mineral Supply Chains in Indonesia**

The United States has been slow to incorporate Indonesia into its critical mineral supply chains. This is largely because companies are sometimes reluctant to invest in Indonesia, especially in politically sensitive sectors like mining. Indonesia's regulatory environment is complex, and the state plays an assertive role in resource management, including



through partial or majority ownership of mining companies. For example, the government recently acquired a controlling interest in the Indonesian operations of foreign-owned mining company Vale in 2024 and secured majority ownership of Freeport Indonesia in 2018. Meanwhile, Chinese firms like Tsingshan and Huayue dominate the nickel sector. These factors create barriers for American investors entering the market despite the strategic value of Indonesia's mineral resources.

From a private-sector perspective, an interventionist state coupled with China's hammerlock on nickel has dimmed Indonesia's attractiveness for foreign investors. However, this evaluation of Indonesia's promise as a partner leaves out strategic considerations. If the goal of the United States is to secure access to critical minerals while reducing dependence on China, Indonesia remains a critical partner to engage, and its downstreaming policies offer clarity on the alignment of both countries' objectives. In recent years the Indonesian state has begun consolidating control over natural resources, including critical minerals, to ensure they are exploited and developed in ways that are more beneficial (or perceived as more beneficial) to the national interest.<sup>10</sup> Rather than closing critical minerals off from foreign investors and overseas markets, this creates a direct channel for engagement as long as strategic interests are aligned.

Two recent developments are important to consider regarding engagement with Indonesia. First, Indonesia has consolidated state-owned mining assets under a single holding company, MIND ID. This includes Aneka Tambang (gold, nickel, and bauxite), Bukit Asam (coal), Inalum (aluminum), PT Timah (tin), and Freeport (copper and gold). This consolidated entity reported assets of \$17.5 billion and revenue of \$8.7 billion in 2024, spanning Indonesia's entire critical mineral landscape from upstream to downstream production.<sup>11</sup> This consolidation enables the state to coordinate resources more effectively and direct them toward national development goals.

Second, in 2025 Indonesia launched Danantara, a state-owned investment fund that functions as a "super holding company," controlling all state enterprises, including MIND ID. One of Danantara's priority investment areas is energy security, including maximizing value from Indonesia's natural resources such as nickel, copper, and coal. It is seeking foreign partners willing not only to invest in these projects but also, and perhaps more importantly, to provide technology, skills, and knowledge.

Foreign firms may be wary of a mining sector with such a highly centralized ownership structure that is deeply entwined with the state. In terms of geopolitics, however, this structure can simplify engagement when broader strategic objectives are aligned. Rather than dealing with a dispersed set of actors and overlapping interests, there is one relatively clear channel for foreign investors and governments that want access to Indonesia's critical minerals: it runs through MIND ID and Danantara. Additionally, their negotiating position is straightforward. Indonesia has the raw materials but needs technology, capital, and expertise to achieve its industrial ambitions and move up global value chains.

## A Critical Minerals Triumvirate in the Asia-Pacific

Foreign partners willing to accommodate Indonesia's interests will find that the country is a potentially attractive alternative for building critical mineral supply chains that do not pass through China. The United States is well positioned to take advantage of this opportunity. The strategic value for Indonesia of expanding participation in global clean energy and semiconductor supply chains is aligned with U.S. efforts to build more secure critical mineral supply chains and de-risk from China. The United States can help construct a production ecosystem that runs through Indonesia by providing research and development support, technology transfers, investment, and upskilling. In so doing, it can

diversify and secure access to critical minerals that are rare or uneconomical to produce domestically.

Other U.S. partners in the region, such as Australia, can play a complementary role in supply chains. Australia has large copper and lithium deposits, which, along with nickel and cobalt, are key components in making lithium-ion batteries. It also has large bauxite, coal, and zinc deposits; is a well-established mining industry; and is a natural trade and investment partner of neighboring Indonesia. Australia is already working to develop germanium and gallium from its own bauxite and zinc resources, an effort that could easily be expanded to include Indonesia.<sup>12</sup> The Australian government, like the United States, is reluctant to intervene directly in economic affairs and influence where private companies invest and how they build their supply chains. But when it comes to critical minerals, there is a strong argument to be made that such decisions should not be left purely to market forces and that geopolitics and strategic interests are important factors.

A critical minerals triumvirate consisting of the United States, Australia, and Indonesia would possess sufficient raw material, technology, and know-how to begin building alternative critical mineral supply chains that steer clear of China. The trio also share high-level strategic alignment, creating opportunities for a critical mineral partnership that could benefit the national interests of all parties. Establishing a high-level strategic framework between the three countries would set the tone for cooperation by defining basic terms and articulating broadly shared objectives and investment goals, similar to the existing U.S.-Australia critical minerals framework. Within this structure, the United States could provide subsidies, tax credits, and offtake guarantees to de-risk private investment from U.S. firms willing to integrate Indonesia and Australia into their value chains.

The Indonesian government already offers a range of incentives, including tax holidays and import

duty exemptions, to foreign companies that invest in its downstream critical mineral sectors.<sup>13</sup> U.S. public finance institutions such as the Export-Import Bank of the United States and the Development Finance Corporation are well positioned to provide additional inducements by underwriting mining, refining, and processing projects based in the region. For the production of germanium and gallium, this could require subsidizing losses, which the U.S. government has historically been reluctant to do. Given the geostrategic value of these products, it may be time to rethink that calculus.

Rather than competing directly with Indonesian mines, Australian mines might be more willing to invest directly in Indonesian upstream extraction and processing technologies if there is a reasonable expectation that a large market like the United States will absorb the output from these ventures. This is where simple mechanisms like offtake guarantees can play an outsized role.

To build more resilient critical mineral supply networks and safeguard the production of next-generation technologies, both the U.S. and Australian governments should focus less on onshoring production and instead explore policies that change private investment incentives and make decentralized supply chains running through trusted partners an attractive option. In exchange for access to its critical minerals, Indonesia will likely seek equity participation, workforce upskilling, and possibly technology transfers. If advanced partners like the United States and Australia are willing to accommodate these expectations, they will find receptive negotiating partners in institutions such as Danantara and MIND ID. Building resilient critical mineral supply chains means that not every country will get everything it wants, but collectively countries will end up in an improved strategic position. A critical minerals triumvirate in the Asia-Pacific based on this understanding can begin creating such an equilibrium. ∞

## Endnotes

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